



Synthesis, antimicrobial and α -glucosidase inhibition of new benzimidazole-1,2,3-triazole-indoline derivatives: a combined experimental and computational venture

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Abstract

In the current research work, a new series of benzimidazole-1,2,3-triazole-indoline derivatives (**5a–r**) were designed and synthesized by employing click reaction between substituted *N*-propargylated benzimidazole derivatives and in situ formed substituted 2-azido-1-(indolin-1-yl)ethanone derivatives in moderate to good yields. The structure of the prepared compounds was confirmed with the help of 1D, 2D-NMR, FTIR and HRMS. The synthesized compounds have been evaluated for their biocidal effects against four bacterial and two fungal strains. The obtained results indicate stronger inhibitory effect of compound **5o** against *E. coli*, while compound **5r** showed good inhibition against all the tested strains except *B. subtilis*. In vitro α -glucosidase inhibition of all synthesized derivatives identified **5p** (IC_{50} value = 0.015 ± 0.0003 $\mu\text{mol/mL}$) and **5r** (IC_{50} value = 0.018 ± 0.0008 $\mu\text{mol/mL}$) as potent inhibitors of α -glucosidase, even better than standard drug Acarbose. Molecular modelling studies into the binding sites of α -glucosidase indicate the stable anchoring of **5p** and **5r** with favourable binding interactions. In addition, pharmacokinetic/dynamic attributes of compounds were estimated by using the ADMET profile.

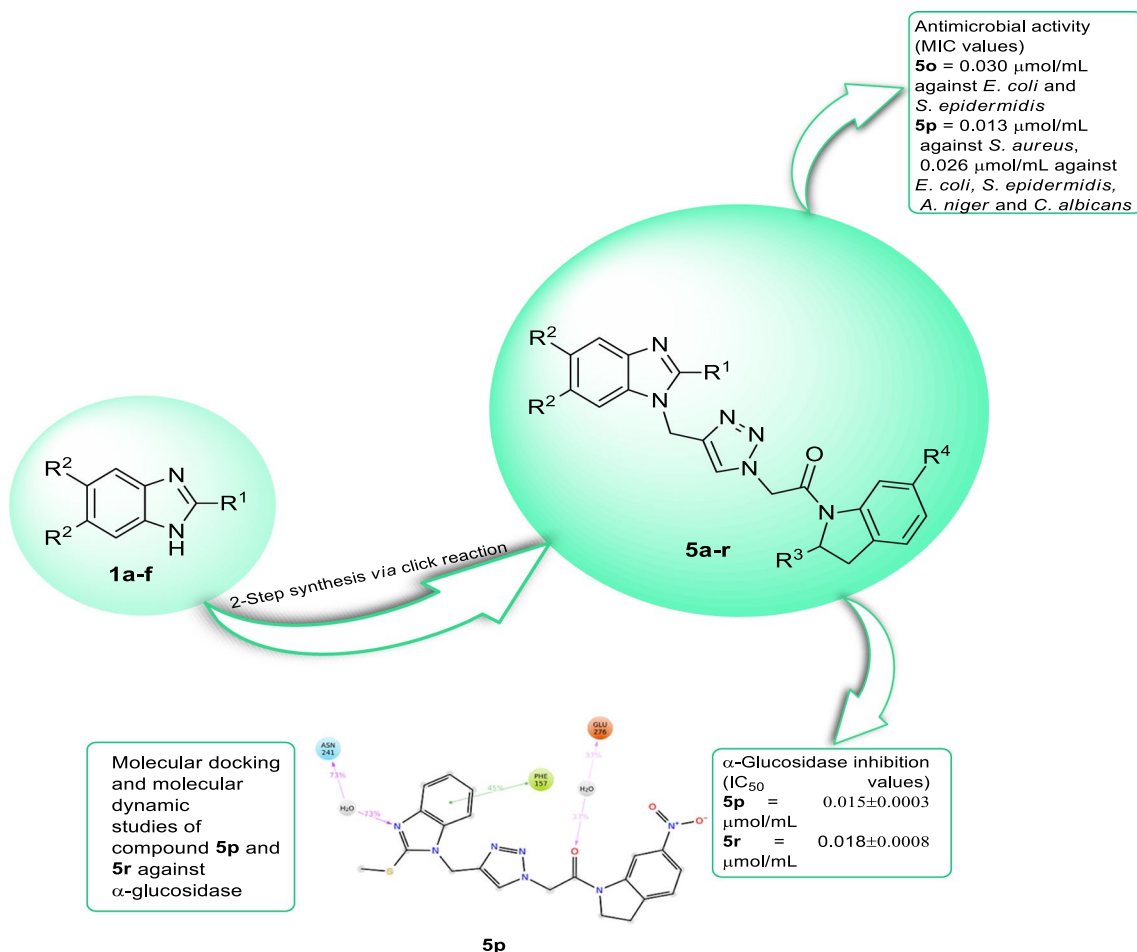
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Graphical abstract



Keywords Benzimidazole · 1,2,3-triazole · Indoline · Antimicrobial · α -glucosidase inhibition · Molecular dynamics

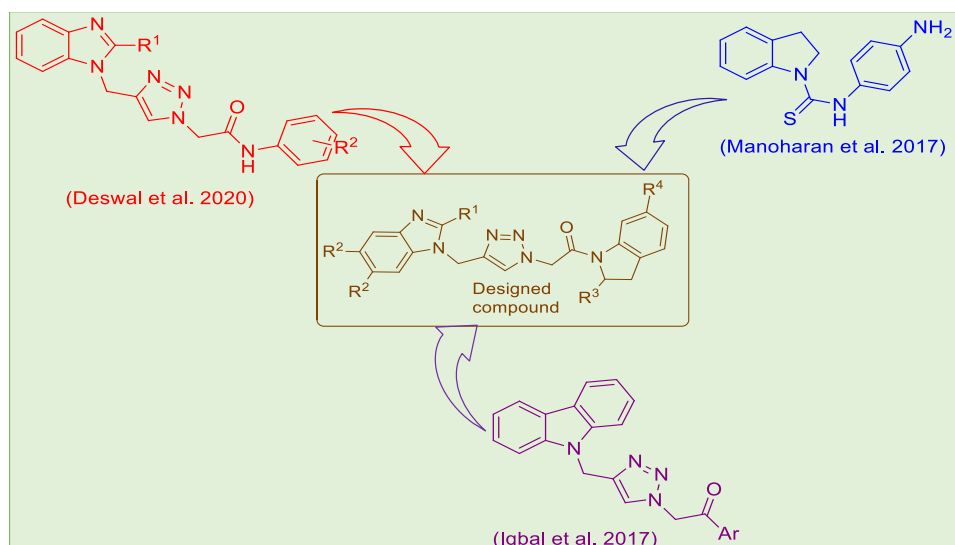
Introduction

Nowadays, the designation of antimicrobial agents has recently piqued the interest of researchers all over the world, owing to the emergence of medication resistance among numerous microorganisms that are responsible for a variety of toxic illnesses and extensive epidemics in humans (B'Bhatt and Sharma, 2017; Punia et al. 2022; Kumar et al. 2022; Kocyigit et al. 2017, 2018; Mor and Khatri, 2022). Excessive use of antimicrobial drugs raises mortality and morbidity rates. Without efficient antimicrobial medications, medical procedures including organ transplantation, diabetes care, cancer chemotherapy and major surgery would become highly unsafe in the future. As a result, there is a continuing demand for new compounds with potential antimicrobial activity (Srinivasan et al. 2019).

Diabetes mellitus is a chronic metabolic disorder characterized by persistent hyperglycaemia, resulting from

defects in insulin secretion or insulin action. The chronic hyperglycaemia is associated with various side effects such as eyes, kidney, nerves, heart and blood vessels damage. The International diabetes federation (IDF, 9th edition) in 2019 estimated that about 463 million people were suffering from diabetes and by the year 2045 more than 700 million people will subsequently have this disease (Saeedi et al. 2019). Type-II diabetes is the most common form of disease and accounts for about 95% of the total diabetic population (Ibrahim et al. 2014). Among the different therapeutic strategies used for treating type-II diabetes, the targeting of carbohydrate hydrolysing enzymes is an important approach for the control of glucose level in the human body (Deswal et al. 2021a, b, c). α -glucosidase present in the epithelium of the small intestine catalyses the hydrolysis of oligosaccharides and disaccharides to monosaccharides such as glucose (Duhan et al. 2021; Gong et al. 2017). Clinically used drugs such as Acarbose, Miglitol and Voglibose are α -glucosidase

Scheme 1 Rational designing of benzimidazole-1,2,3-triazole-indoline derivatives



inhibitors, but due to their adverse side effects, there is a need to explore and synthesize some new potent compounds for the treatment of type-II diabetes mellitus (Zawawi et al. 2016). Targeting of an α -glucosidase enzyme is an effective strategy for treatment of diabetes. Therefore, developing novel and powerful α -glucosidase inhibitors is highly tough for medicinal chemistry researchers.

The design and the synthesis of new bio-relevant nitrogen containing heterocycles are at the forefront of the drug discovery process. Benzimidazole is an important heterocyclic scaffold and many naturally occurring compounds incorporate them as a part of their structural framework (Tahlan et al. 2019). From the structural standpoint, benzimidazole can be considered as cyclic analogue of amidines. A wide variety of benzimidazole derivatives have been examined in vitro and in vivo to discover new medicinal molecules (Law and Yeong 2021). Recently, benzimidazole-based derivatives have been investigated as potent antimicrobial and antidiabetic agents (Marinescu 2021; Aroua et al. 2021). Moreover, literature reports reveal that 1,2,3-triazole derivatives are endowed with various therapeutic activities such as antimicrobial, antidiabetic, anticancer, anti-HIV, antituberculosis, etc. (Iqbal et al. 2017; Dheer et al. 2017). Triazole moiety has gained exceptional recognition in the drug discovery process because the drugs bearing triazole moiety, i.e. Tazobactam, Cefatrizine, Carboxyamidotriazole, etc., are in clinical use for treatment of various diseases. Furthermore, indoline skeleton also emerges as a key structural unit of many alkaloids, pharmaceutical and biologically active compounds (Manoharan et al. 2017; Chang et al. 2010; Al-Shareef et al. 2016). Indoline and its derivatives offer high degree of structural diversity which proves their usefulness in search of new leads.

The aim of present research work is to use a molecular hybridization strategy to design (Scheme 1) and synthesize

new benzimidazole-1,2,3-triazole-indoline derivatives (**5a–r**). Various spectroscopic approaches were used to successfully characterize the synthesized derivatives. In silico ADMET and drug-likeness score of the prepared compounds were also checked to evaluate their bioavailability and toxicity. The antimicrobial and α -glucosidase inhibition profiles of the produced compounds were evaluated using in vitro tests. Further, molecular docking and molecular dynamics studies of two hit compounds were performed to investigate their inner actions with α -glucosidase.

Experimental

Materials and methods

The melting point of benzimidazole-1,2,3-triazoles-indoline hybrids was determined by Decibel Melting Point apparatus and reported uncorrected. Chemicals used were purchased from Sigma-Aldrich, Alfa-Aesar, HiMedia and used as such in the experiment without any further purification. SHIMADZU IR AFFINITY-I FT-IR is used for recording infrared spectra (ν in cm^{-1}) by using KBr matrix and Bruker Avance III (400 MHz) was utilized for taking 1D and 2D NMR spectra (δ in ppm). SCIEX TripleTOF 5600 and 5600+/SCIEX spectrometer was employed for HRMS and values were recorded in the form of m/z . α -glucosidase from *Saccharomyces cerevisiae* has been purchased from Sigma-Aldrich for in vitro studies.

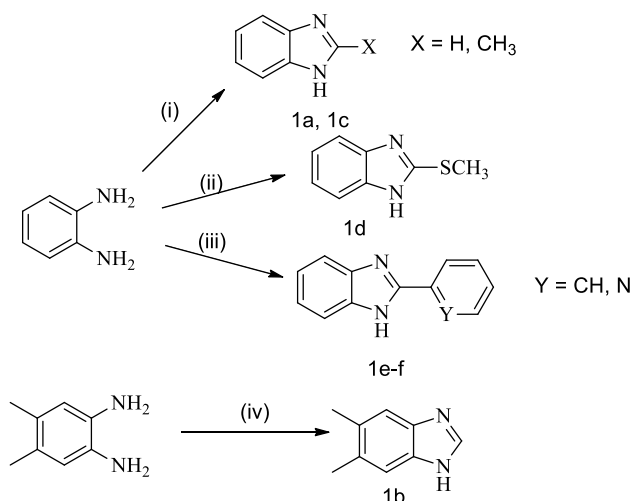
General method for the synthesis of substituted benzimidazoles (1a–f)

Substituted benzimidazoles (Scheme 2) were prepared by reported procedures (Deswal et al. 2021a, b, c; Wagner and Millett 1943).

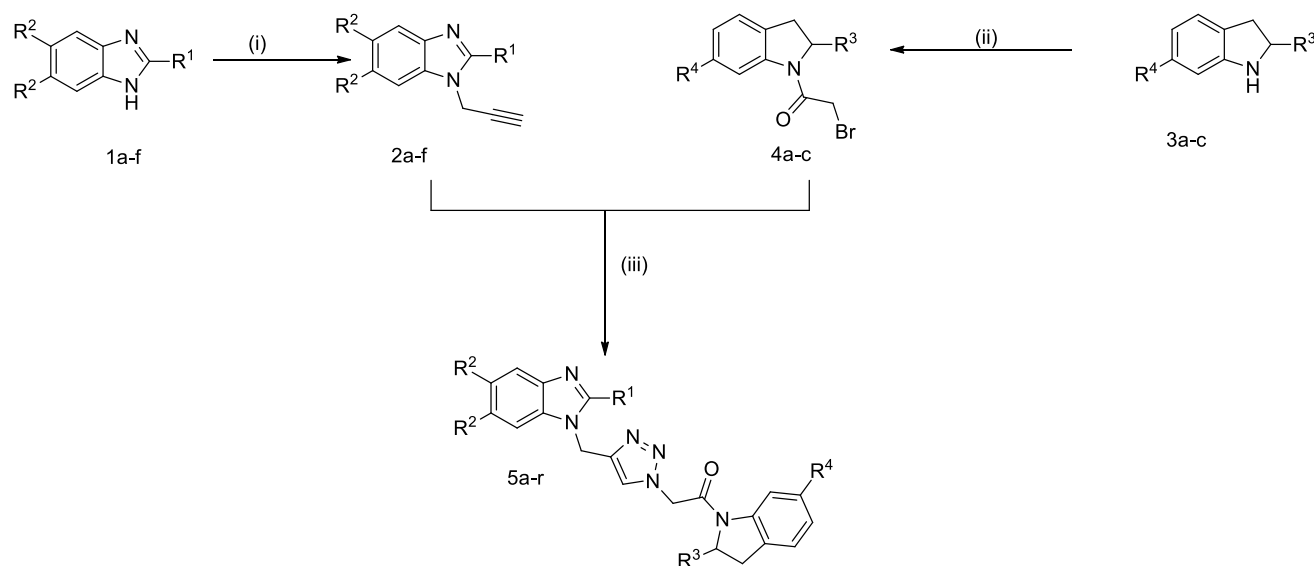
Reagents and conditions: (i) formic acid/acetic acid, reflux, 4–5 h (ii) (a) CS₂, KOH, ethanol; (b) CH₃I, ethanol (iii) benzaldehyde/2-pyridinecarboxaldehyde, H₂O₂, HCl, H₂O, 100 °C, stirring, 2–3 h (iv) formic acid, reflux, 4–5 h.

General method for the synthesis of *N*-propargylbenzimidazoles (2a–f)

To a stirred solution of substituted benzimidazoles (2 mmol) in dimethyl formamide (10 mL) in the presence of anhydrous potassium carbonate (3 mmol), propargyl bromide (2 mmol) was added dropwise and the reaction mixture was stirred at 0–15 °C temperature for 4–5 h. Progress of the reaction was



Scheme 2 Route for the synthesis of compounds **1a–f**



Scheme 3 Route for the synthesis of compounds **2a–f**, **4a–c** and **5a–r**

checked by TLC. After completion of the reaction, reaction mixture was added to ice-cold water and the compounds were extracted using ethyl acetate. The excess of solvent was evaporated under vacuum to get the desired products (Deswal et al. 2020). The solid residues were purified by column chromatography using ethyl acetate/hexane (3:7, v: v) as eluent to afford substituted-1-(prop-2-yn-1-yl)-1*H*-benzo[*d*]imidazole (**2a–f**) (Scheme 3).

General methods for the synthesis of 2-bromo-1-(indolin-1-yl)ethanone derivatives (4a–c)

To a stirred solution of substituted indolines (2 mmol) and anhydrous potassium carbonate (3 mmol), bromoacetyl bromide (2 mmol) was added dropwise (dissolved in DCM) and the reaction was further stirred for 1–3 h at 10–15 °C. After the reaction completion, ice-cold water was added to the reaction mixture and the organic layer separates out (Hecker et al., 2019). The solvent was evaporated and the compounds were recrystallized in chloroform (Scheme 3).

General method for the synthesis of benzimidazole-1,2,3-triazole-indoline hybrids (5a–r)

To a round bottom flask, substituted 2-bromo-1-(indolin-1-yl)ethanone derivatives (2 mmol) were dissolved into DMF and added to it sodium azide (6 mmol, aqueous solution) with continuous stirring for about half an hour. Then, substituted-1-(prop-2-yn-1-yl)-1*H*-benzo[*d*]imidazole (2 mmol) was added followed by addition of CuSO₄·5H₂O (10 mol %) and sodium ascorbate (20 mol%). The reaction mixture was allowed for continuous stirring for 4–6 h at

40–55 °C (Chauhan et al. 2019; Punia et al. 2021a, b). The reaction progress was checked by TLC. After completion of the reaction, the reaction was quenched with the addition of ice-cold aqueous ammonia solution and the compound was extracted by ethyl acetate. After extraction, the ethyl acetate layer was dried over anhydrous sodium sulphate and the solvent was evaporated by rotary evaporator under vacuum. The synthesized compounds were recrystallized by using ethanol. The desired products benzimidazole-1,2,3-triazole-indoline derivatives (Table 1, **5a–r**) were isolated in considerable to good yield (Scheme 3).

Reagents and conditions: (i) K_2CO_3 , propargyl bromide, DMF, 0–15 °C, stirring 4–5 h (ii) K_2CO_3 , bromoacetyl bromide, toluene, 10–15 °C, stirring 1–3 h (iii) NaN_3 , $CuSO_4 \cdot 5H_2O$, sodium ascorbate, DMF: H_2O (8:2), 50 °C, stirring 5–6 h.

2-(4-((1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(indolin-1-yl)ethanone (5a)

Brown colour, yield: 75%, M.P.: 250–252 °C, 1H NMR (400 MHz, DMSO) δ 8.68 (s, 1H), 8.31 (s, 1H), 8.07 (s, 1H), 7.97 (d, $J=4$ Hz, 1H), 7.94 (d, $J=4$ Hz, 1H), 7.70–7.65 (m, 1H), 7.53 (d, $J=8.2$ Hz, 3H), 7.29–7.19 (m, 1H), 5.53 (s, 2H), 5.40 (s, 2H), 4.09 (t, $J=8.5$ Hz, 2H), 1.36 (t, $J=8.6$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 163.46, 148.96, 145.29, 144.41, 143.88, 142.75, 140.76, 136.14, 125.93, 125.58, 122.72, 122.09, 119.94, 110.70, 52.45, 44.96, 39.19, 29.38. IR (KBr, ν_{max}): 3159 (C–H str., triazole), 2965 (C–H

str., aromatic), 1678 (C=O str. amide), 1479, 1441, 1406 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{20}H_{19}N_6O$ $[M+H]^+$ 359.1615; found 359.1622.

2-(4-((5,6-dimethyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(indolin-1-yl)ethanone (5b)

Brown colour, yield: 77%, M.P.: 140–145 °C, 1H NMR (400 MHz, DMSO) δ 8.08 (s, 1H), 7.96 (s, 1H), 7.94 (s, 1H), 7.29 (s, 1H), 7.27 (s, 1H), 7.15 (t, $J=7.0$ Hz, 2H), 7.03 (t, $J=7.0$ Hz, 2H), 5.71 (s, 2H), 5.54 (s, 2H), 4.19 (t, $J=8.1$ Hz, 2H), 3.23–3.18 (t, $J=8.0$ Hz, 2H), 2.36 (s, 3H), 2.27 (s, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.49, 142.89, 142.26, 132.20, 127.56, 125.95, 125.47, 124.73, 124.42, 124.36, 116.23, 52.49, 46.86, 39.29, 28.01, 20.54, 20.44. IR (KBr, ν_{max}): 3131 (C–H str., triazole), 2942 (C–H str., aromatic), 1668 (C=O str. amide), 1484, 1436, 1406 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{22}H_{23}N_6O$ $[M+H]^+$ 387.1889; found 387.1895.

1-(indolin-1-yl)-2-(4-((2-methyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5c)

Off white colour, yield: 78%, M.P.: 198–200 °C, 1H NMR (400 MHz, $CDCl_3$) δ 8.56 (d, $J=8.2$ Hz, 2H), 8.11 (d, $J=8.0$ Hz, 1H), 7.91–7.88 (m, 1H), 7.84 (t, $J=7.8$ Hz, 1H), 7.40 (d, $J=4.8$ Hz, 1H), 7.36 (s, 1H), 7.25 (d, $J=7.3$ Hz, 1H), 7.15 (t, $J=7.7$ Hz, 1H), 7.05 (t, $J=7.3$ Hz, 1H), 5.85 (s, 2H), 5.47 (s, 2H), 4.33 (t, $J=8.3$ Hz, 2H), 3.36 (t, $J=8.2$ Hz, 2H), 1.68 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 165.14, 150.56, 148.18, 142.55, 137.31, 137.01, 127.70, 124.53, 124.04, 123.87, 123.78, 122.89, 120.39, 117.10, 109.59, 48.59, 47.14, 40.07, 28.42, 14.59. IR (KBr, ν_{max}): 3134 (C–H str., triazole), 2944 (C–H str., aromatic), 1666 (C=O str. amide), 1461, 1437, 1406 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{21}H_{21}N_6O$ $[M+H]^+$ 373.1771; found 373.1774.

1-(indolin-1-yl)-2-(4-((2-(methylthio)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5d)

Off white colour, yield: 83%, M.P.: 195–197 °C, 1H NMR (400 MHz, DMSO) δ 8.06 (s, 1H), 7.94 (d, $J=7.9$ Hz, 1H), 7.61 (dd, $J=6.5, 2.4$ Hz, 1H), 7.56 (dd, $J=6.4, 2.5$ Hz, 1H), 7.27 (t, $J=7.3$ Hz, 1H), 7.21–7.16 (m, 2H), 7.14 (d, $J=7.4$ Hz, 1H), 7.03 (t, $J=7.8$ Hz, 1H), 5.52 (s, 2H), 5.48 (s, 2H), 4.18 (t, $J=8.5$ Hz, 2H), 3.20 (t, $J=8.4$ Hz, 2H), 2.74 (s, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.45, 152.71, 143.46, 142.87, 142.28, 136.56, 132.18, 127.56, 125.59, 125.45, 124.35, 122.08, 118.00, 116.28, 110.30, 52.44, 46.87, 39.19, 28.00, 14.90. IR (KBr, ν_{max}): 3132 (C–H str., triazole), 2933 (C–H str., aromatic), 1675 (C=O str. amide), 1482, 1440, 1405 (C=C str., aromatic) cm^{-1} . HRMS (ESI):

Table 1 Substituents present in compounds **5a–r**

Comp. No	R ¹	R ²	R ³	R ⁴
5a	H	H	H	H
5b	H	CH ₃	H	H
5c	CH ₃	H	H	H
5d	SCH ₃	H	H	H
5e	C ₆ H ₅	H	H	H
5f	C ₅ H ₄ N	H	H	H
5g	H	H	CH ₃	H
5h	H	CH ₃	CH ₃	H
5i	CH ₃	H	CH ₃	H
5j	SCH ₃	H	CH ₃	H
5k	C ₆ H ₅	H	CH ₃	H
5l	C ₅ H ₄ N	H	CH ₃	H
5m	H	H	H	NO ₂
5n	H	CH ₃	H	NO ₂
5o	CH ₃	H	H	NO ₂
5p	SCH ₃	H	H	NO ₂
5q	C ₆ H ₅	H	H	NO ₂
5r	C ₅ H ₄ N	H	H	NO ₂

C₅H₄N is 2-pyridinyl substituent

m/z calculated for $C_{21}H_{21}N_6OS$ $[M+H]^+$ 405.1492; found 405.1405.

1-(indolin-1-yl)-2-(4-((2-phenyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5e)

Off white colour, yield: 82%, M.P.: 235–237 °C, 1H NMR (400 MHz, DMSO) δ 8.11 (s, 1H), 8.03–7.97 (m, 2H), 7.95 (d, $J=8.0$ Hz, 1H), 7.71 (d, $J=8.7$ Hz, 1H), 7.68 (d, $J=8.6$ Hz, 1H), 7.61 (s, 1H), 7.61 (s, 2H), 7.34–7.21 (m, 3H), 7.16 (t, $J=7.6$ Hz, 1H), 7.04 (t, $J=7.3$ Hz, 1H), 5.60 (s, 2H), 5.55 (s, 2H), 4.19 (t, $J=8.4$ Hz, 2H), 3.21 (t, $J=8.3$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 164.47, 143.13, 142.86, 132.18, 130.57, 130.32, 129.96, 129.23, 127.58, 125.78, 125.46, 124.35, 123.01, 122.67, 119.64, 116.27, 111.65, 52.53, 46.86, 39.24, 28.01. IR (KBr, ν_{max}): 3138 (C–H str., triazole), 3062 (C–H str., aromatic), 1669 (C=O str. amide), 1479, 1459, 1440 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{26}H_{23}N_6O$ $[M+H]^+$ 435.1928; found 435.1930.

1-(indolin-1-yl)-2-(4-((2-(pyridin-2-yl)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5f)

Light brown colour, yield: 74%, M.P.: 238–240 °C, 1H NMR (400 MHz, DMSO) δ 8.82 (s, 1H), 8.39 (d, $J=7.3$ Hz, 1H), 8.05 (d, $J=7.5$ Hz, 1H), 8.01 (s, 1H), 7.91 (d, $J=7.8$ Hz, 1H), 7.84 (d, $J=7.5$ Hz, 1H), 7.75 (d, $J=7.7$ Hz, 1H), 7.57 (d, $J=5.1$ Hz, 1H), 7.36 (t, $J=7.5$ Hz, 1H), 7.28 (dd, $J=16.1, 7.5$ Hz, 2H), 7.14 (t, $J=7.3$ Hz, 1H), 7.02 (t, $J=7.1$ Hz, 1H), 6.26 (s, 2H), 5.47 (s, 2H), 4.15 (t, $J=8.1$ Hz, 2H), 3.18 (t, $J=7.9$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 164.51, 150.44, 149.32, 143.61, 142.88, 138.02, 132.14, 127.55, 125.59, 125.43, 124.81, 124.30, 123.81, 123.05, 119.99, 116.25, 112.01, 52.39, 46.83, 40.86, 27.98. IR (KBr, ν_{max}): 3143 (C–H str., triazole), 3046 (C–H str., aromatic), 1675 (C=O str. amide), 1481, 1442, 1404 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{25}H_{22}N_7O$ $[M+H]^+$ 436.1880; found 436.1888.

2-(4-((1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(2-methylindolin-1-yl)ethanone (5g)

Dark red colour, yield: 81%, M.P.: 65–67 °C, 1H NMR (400 MHz, DMSO) δ 8.37 (s, 1H), 8.17 (s, 1H), 7.89 (d, $J=7.7$ Hz, 1H), 7.68 (t, $J=8.1$ Hz, 2H), 7.34–7.27 (m, 2H), 7.23 (dd, $J=6.5, 6.0$ Hz, 1H), 7.19–7.14 (m, 1H), 7.06 (d, $J=7.3$ Hz, 1H), 5.75 (d, $J=16.8$ Hz, 1H), 5.63 (s, 2H), 5.48 (d, $J=16.8$ Hz, 1H), 4.80 (t, $J=14.1$ Hz, 1H), 2.71 (d, $J=16.9$ Hz, 1H), 1.29 (d, $J=5.7$ Hz, 3H), 1.19 (d, $J=6.4$ Hz, 1H). ^{13}C NMR (100 MHz, DMSO) δ 192.88, 151.35, 147.91, 144.46, 142.42, 137.00, 134.06, 130.90,

125.50, 124.88, 122.94, 120.91, 119.75, 119.29, 111.09, 64.86, 54.56, 43.71, 33.09, 15.02. IR (KBr, ν_{max}): 3178 (C–H str., triazole), 2921 (C–H str., aromatic), 1663 (C=O str. amide), 1484, 1458, 1407 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{21}H_{21}N_6O$ $[M+H]^+$ 373.1771; found 373.1780.

2-(4-((5,6-dimethyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(2-methylindolin-1-yl)ethanone (5h)

Light grey colour, yield: 77%, M.P.: 75–77 °C, 1H NMR (400 MHz, DMSO) δ 8.23 (s, 1H), 7.89 (d, $J=6.1$ Hz, 1H), 7.49–7.07 (m, 4H), 5.95 (s, 2H), 5.77 (d, $J=16.3$ Hz, 1H), 5.51 (d, $J=16.3$ Hz, 1H), 4.55–4.46 (m, 1H), 2.72 (d, $J=15.6$ Hz, 2H), 2.41 (s, 3H), 2.28 (s, 3H), 1.31 (d, $J=12$ Hz, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.00, 141.45, 141.25, 138.95, 137.04, 134.73, 131.32, 129.92, 127.57, 126.60, 125.88, 124.75, 117.32, 111.05, 55.05, 52.02, 36.61, 29.76, 21.87, 20.50, 14.65. IR (KBr, ν_{max}): 3133 (C–H str., triazole), 2969 (C–H str., aromatic), 1662 (C=O str. amide), 1480, 1434, 1408 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{23}H_{25}N_6O$ $[M+H]^+$ 401.2084; found 401.2092.

2-(4-((2-methyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(2-methylindolin-1-yl)ethanone (5i)

Off white colour, yield: 73%, M.P.: 235–237 °C, 1H NMR (400 MHz, DMSO) δ 8.12 (s, 1H), 7.88 (d, $J=7.8$ Hz, 1H), 7.59 (dd, $J=22.6, 8.3$ Hz, 2H), 7.31 (d, $J=7.1$ Hz, 1H), 7.24–7.11 (m, 3H), 7.06 (t, $J=7.3$ Hz, 1H), 5.73 (d, $J=16.8$ Hz, 1H), 5.48 (s, 2H), 5.22 (d, $J=15.3$ Hz, 1H), 5.00–4.66 (m, 1H), 3.42 (d, 1H), 2.68 (d, $J=15.8$ Hz, 1H), 2.31 (s, 3H), 1.28 (d, $J=5.8$ Hz, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.03, 146.20, 143.47, 142.23, 141.43, 136.56, 131.31, 127.57, 125.77, 124.71, 122.03, 118.00, 117.33, 111.11, 55.01, 51.90, 39.21, 36.58, 21.78, 14.11. IR (KBr, ν_{max}): 3157 (C–H str., triazole), 2934 (C–H str., aromatic), 1661 (C=O str. amide), 1485, 1440, 1401 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $C_{22}H_{23}N_6O$ $[M+H]^+$ 387.1928; found 387.1931.

1-(2-methylindolin-1-yl)-2-(4-((2-(methylthio)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5j)

Light brown colour, yield: 70%, M.P.: 170–172 °C, 1H NMR (400 MHz, DMSO) δ 8.12 (s, 1H), 7.88 (d, $J=7.8$ Hz, 1H), 7.62 (d, $J=8.1$ Hz, 1H), 7.56 (d, $J=8.5$ Hz, 1H), 7.31 (d, $J=7.1$ Hz, 1H), 7.24–7.11 (m, 3H), 7.06 (t, $J=7.3$ Hz, 1H), 5.73 (d, $J=16.8$ Hz, 1H), 5.48 (s, 3H), 5.22 (d, $J=16.8$ Hz, 1H), 4.79 (dd, $J=12.9, 6.2$ Hz, 1H), 3.47–3.38 (m, 1H), 2.75

(s, 3H), 1.28 (d, $J=5.8$ Hz, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.03, 152.69, 143.47, 142.23, 141.43, 136.56, 131.31, 127.57, 125.77, 124.71, 122.03, 118.00, 117.33, 110.30, 55.01, 51.90, 39.21, 36.58, 21.78, 14.89. IR (KBr, ν_{max}): 3130 (C–H str., triazole), 2928 (C–H str., aromatic), 1656 (C=O str. amide), 1441, 1404 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{22}\text{H}_{23}\text{N}_6\text{OS}$ $[\text{M} + \text{H}]^+$ 419.1649; found 419.1654.

1-(2-methylindolin-1-yl)-2-(4-((2-phenyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5k)

Light brown colour, yield: 69%, M.P.: 100–102 °C, ^1H NMR (400 MHz, DMSO) δ 8.82 (s, 1H), 8.39 (d, $J=6.7$ Hz, 1H), 8.12–7.97 (m, 2H), 7.84 (t, $J=8.7$ Hz, 2H), 7.75 (d, $J=7.8$ Hz, 1H), 7.57 (d, $J=4.5$ Hz, 1H), 7.35 (t, $J=7.5$ Hz, 1H), 7.30 (dd, $J=6.7$, 3.9 Hz, 2H), 7.16 (t, $J=7.6$ Hz, 1H), 7.05 (t, $J=7.1$ Hz, 1H), 6.26 (s, 2H), 5.68 (d, $J=17.0$ Hz, 1H), 5.41 (d, $J=16.8$ Hz, 1H), 4.79–4.72 (m, 1H), 2.69 (d, $J=15.9$ Hz, 1H), 1.25 (d, $J=5.9$ Hz, 3H), 1.19 (d, $J=6.3$ Hz, 1H). ^{13}C NMR (100 MHz, DMSO) δ 164.06, 149.37, 143.58, 141.43, 138.03, 131.30, 127.56, 125.83, 125.78, 125.00, 124.80, 124.69, 123.83, 123.06, 120.02, 117.33, 117.31, 112.00, 54.97, 51.85, 44.28, 36.57, 21.77. IR (KBr, ν_{max}): 3148 (C–H str., triazole), 2973 (C–H str., aromatic), 1666 (C=O str. amide), 1482, 1441, 1406 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{27}\text{H}_{25}\text{N}_6\text{O}$ $[\text{M} + \text{H}]^+$ 450.2084; found 450.2089.

1-(2-methylindolin-1-yl)-2-(4-((2-(pyridin-2-yl)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5l)

Light brown colour, yield: 71%, M.P.: 95–97 °C, ^1H NMR (400 MHz, DMSO) δ 8.82 (s, 1H), 8.39 (d, $J=8.0$ Hz, 1H), 8.19–7.95 (m, 2H), 7.84 (t, $J=7.5$ Hz, 2H), 7.75 (d, $J=8.1$ Hz, 1H), 7.62–7.49 (m, 1H), 7.35 (t, $J=7.4$ Hz, 2H), 7.29 (s, 1H), 7.16 (t, $J=6.6$ Hz, 1H), 7.05 (t, $J=7.2$ Hz, 1H), 6.26 (s, 2H), 5.68 (d, $J=17.2$ Hz, 1H), 5.41 (d, $J=16.5$ Hz, 1H), 4.74 (dd, $J=14.7$, 5.9 Hz, 1H), 2.70 (d, $J=15.8$ Hz, 2H), 1.25 (d, $J=5.6$ Hz, 3H). ^{13}C NMR (100 MHz, DMSO) δ 164.02, 149.34, 143.56, 138.03, 127.58, 125.75, 124.84, 124.72, 123.85, 123.10, 121.37, 119.95, 117.33, 111.97, 55.06, 51.80, 40.90, 36.56, 21.76. IR (KBr, ν_{max}): 3153 (C–H str., triazole), 2971 (C–H str., aromatic), 1667 (C=O str. amide), 1476, 1442, 1408 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{26}\text{H}_{24}\text{N}_7\text{O}$ $[\text{M} + \text{H}]^+$ 450.2037; found 450.2045.

2-(4-((1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(6-nitroindolin-1-yl)ethanone (5m)

Yellow colour, yield: 70%, M.P.: 100–102 °C, ^1H NMR (400 MHz, DMSO) δ 8.80 (s, 1H), 8.36 (s, 1H), 8.13 (s,

1H), 7.97 (d, $J=2.2$ Hz, 1H), 7.94 (s, 1H), 7.68 (dd, $J=11.8$, 7.8 Hz, 1H), 7.54 (s, 1H), 7.52 (s, 1H), 7.29–7.19 (m, 1H), 5.64 (s, 2H), 5.60 (s, 2H), 4.13 (t, $J=8.5$ Hz, 2H), 3.29 (t, $J=8.6$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 167.51, 147.42, 144.41, 143.88, 142.99, 140.76, 126.07, 125.75, 122.84, 122.09, 119.90, 119.81, 111.21, 110.30, 51.37, 47.44, 39.19, 28.13. IR (KBr, ν_{max}): 3140 (C–H str., triazole), 2921 (C–H str., aromatic), 1673 (C=O str. amide), 1484, 1431, 1408 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{20}\text{H}_{18}\text{N}_7\text{O}_3$ $[\text{M} + \text{H}]^+$ 404.1466; found 404.1470.

2-(4-((5,6-dimethyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(6-nitroindolin-1-yl)ethanone (5n)

Light brown colour, yield: 84%, M.P.: 245–247 °C, ^1H NMR (400 MHz, DMSO) δ 8.69 (d, $J=2.0$ Hz, 1H), 8.11 (s, 1H), 7.96 (dd, $J=8.2$, 2.2 Hz, 2H), 7.54 (d, $J=8.2$ Hz, 2H), 7.32–7.00 (m, 1H), 5.60 (s, 4H), 4.31 (t, $J=8.5$ Hz, 2H), 3.32 (s, 2H), 2.34 (s, 3H), 2.29 (s, 3H). ^{13}C NMR (100 MHz, DMSO) δ 165.69, 147.36, 143.79, 140.84, 135.78, 132.01, 126.14, 125.85, 123.75, 122.02, 119.97, 110.33, 52.45, 47.67, 39.21, 28.18, 20.54, 20.41. IR (KBr, ν_{max}): 3161 (C–H str., triazole), 2968 (C–H str., aromatic), 1662 (C=O str. amide), 1473, 1435, 1403 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{22}\text{H}_{22}\text{N}_7\text{O}_3$ $[\text{M} + \text{H}]^+$ 432.1779; found 432.1783.

2-(4-((2-methyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(6-nitroindolin-1-yl)ethanone (5o)

Off white colour, yield: 86%, M.P.: 230–232 °C, ^1H NMR (400 MHz, CDCl_3) δ 8.56 (d, $J=8.2$ Hz, 2H), 8.11 (d, $J=8.0$ Hz, 1H), 7.91–7.88 (m, 1H), 7.84 (t, $J=7.8$ Hz, 1H), 7.40 (d, $J=4.8$ Hz, 1H), 7.36 (s, 1H), 7.25 (d, $J=7.3$ Hz, 1H), 7.15 (t, $J=7.7$ Hz, 1H), 7.05 (t, $J=7.3$ Hz, 1H), 5.85 (s, 2H), 5.47 (s, 2H), 4.33 (t, $J=8.3$ Hz, 2H), 3.36 (t, $J=8.2$ Hz, 2H), 1.68 (s, 3H). ^{13}C NMR (100 MHz, CDCl_3) δ 165.14, 150.56, 148.18, 142.55, 137.31, 137.01, 127.70, 124.53, 124.04, 123.87, 123.78, 122.89, 120.39, 117.10, 109.59, 48.59, 47.14, 39.81, 28.42, 14.23. IR (KBr, ν_{max}): 3128 (C–H str., triazole), 2929 (C–H str., aromatic), 1676 (C=O str. amide), 1485, 1444, 1410 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{21}\text{H}_{20}\text{N}_7\text{O}_3$ $[\text{M} + \text{H}]^+$ 418.1622; found 418.1625.

2-(4-((2-(methylthio)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)-1-(6-nitroindolin-1-yl)ethanone (5p)

Light brown colour, yield: 80%, M.P.: 232–235 °C, ^1H NMR (400 MHz, DMSO) δ 8.68 (s, 1H), 8.12 (s, 1H), 7.96 (d,

$J=3.5$ Hz, 1H), 7.70 (s, 1H), 7.55 (s, 2H), 7.31–7.02 (m, 2H), 5.59 (s, 4H), 4.32 (d, $J=8.8$ Hz, 2H), 4.17 (t, $J=8$ Hz, 2H), 2.68 (s, 3H). ^{13}C NMR (100 MHz, DMSO) δ 165.67, 147.38, 143.78, 142.99, 140.83, 128.69, 126.14, 126.08, 125.53, 122.07, 121.83, 119.96, 119.82, 118.69, 110.33, 52.42, 51.36, 47.70, 47.43, 28.18. IR (KBr, ν_{max}): 3138 (C–H str., triazole), 2951 (C–H str., aromatic), 1696 (C=O str. amide), 1482, 1434, 1408 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{21}\text{H}_{20}\text{N}_7\text{O}_3\text{S}$ $[\text{M} + \text{H}]^+$ 450.1343; found 450.1345.

1-(6-nitroindolin-1-yl)-2-(4-((2-phenyl-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5q)

Light green colour, yield: 85%, M.P.: 240–242 °C, ^1H NMR (400 MHz, DMSO) δ 9.25 (d, $J=8.0$ Hz, 1H), 8.82 (s, 1H), 8.04–7.95 (m, 2H), 7.84 (t, $J=7.5$ Hz, 2H), 7.75 (d, $J=8.1$ Hz, 1H), 7.62–7.49 (m, 1H), 7.35 (t, $J=7.4$ Hz, 2H), 7.29 (s, 1H), 7.16 (t, $J=6.6$ Hz, 1H), 7.05 (t, $J=7.2$ Hz, 1H), 6.26 (s, 2H), 5.55 (s, 1H), 4.77–4.72 (m, 1H), 2.80 (d, $J=15.8$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 164.01, 142.91, 142.67, 135.37, 131.88, 129.25, 128.33, 125.53, 122.01, 121.80, 121.28, 118.62, 114.47, 110.51, 55.64, 47.69, 38.58, 29.92. IR (KBr, ν_{max}): 3142 (C–H str., triazole), 2989 (C–H str., aromatic), 1673 (C=O str. amide), 1479, 1434, 1405 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{26}\text{H}_{22}\text{N}_7\text{O}_3$ $[\text{M} + \text{H}]^+$ 480.1779; found 480.1781.

1-(6-nitroindolin-1-yl)-2-(4-((2-(pyridin-2-yl)-1H-benzo[d]imidazol-1-yl)methyl)-1H-1,2,3-triazol-1-yl)ethanone (5r)

Light brown colour, yield: 81%, M.P.: 250–252 °C, ^1H NMR (400 MHz, DMSO) δ 8.81 (s, 1H), 8.65 (s, 1H), 8.39 (d, $J=7.4$ Hz, 1H), 8.09–7.98 (m, 2H), 7.93 (d, $J=8.2$ Hz, 1H), 7.84 (d, $J=7.6$ Hz, 1H), 7.75 (d, $J=7.7$ Hz, 1H), 7.56 (s, 1H), 7.51 (d, $J=8.2$ Hz, 1H), 7.36 (t, $J=7.5$ Hz, 1H), 7.30 (t, $J=7.4$ Hz, 1H), 6.27 (s, 2H), 5.53 (s, 2H), 4.26 (t, $J=8.4$ Hz, 2H), 3.31 (t, $J=8.4$ Hz, 2H). ^{13}C NMR (100 MHz, DMSO) δ 165.65, 149.29, 147.39, 143.78, 142.59, 140.76, 137.99, 126.07, 125.54, 124.81, 123.82, 123.05, 119.90, 111.98, 110.35, 52.37, 47.68, 40.87, 28.16. IR (KBr, ν_{max}): 3167 (C–H str., triazole), 2948 (C–H str., aromatic), 1657 (C=O str. amide), 1466, 1440, 1403 (C=C str., aromatic) cm^{-1} . HRMS (ESI): m/z calculated for $\text{C}_{25}\text{H}_{21}\text{N}_8\text{O}_3$ $[\text{M} + \text{H}]^+$ 481.1731; found 481.1735.

Antimicrobial assay

The synthesized compounds (**5a–r**) have been tested for their in vitro antimicrobial activity using assay described by Deswal et al. (2021a), Saroya et al. (2022) and Punia et al. (2021a, b).

α -Glucosidase inhibition

All the synthesized compounds (**5a–r**) were examined for their in vitro inhibitory activity against α -glucosidase using method reported by (Xiao and Hogger 2015)s.

Homology model building and validation for compound 5p, 5r

Lack of three-dimensional structure of yeast α -glucosidase prompted us to build a homology model using RaptorX web server (Källberg et al. 2012) using oligo-1,6-glucosidase (PDB ID: 3A47, Gene ID: IMA1) of *Saccharomyces cerevisiae* as a template structure. The prepared model was validated thoroughly. Firstly, Ramachandran plot was obtained by using PROCHECK server (<https://servicesn.mbi.ucla.edu/PROCHECK/>) which supported the strength of model as 92.1% residues were in most favoured regions, 7.2% of residues were in additional allowed region, 0.6% of residues were in generously allowed region, and only 0.2% of residues fell in disallowed regions. Further, Qualitative Model Energy Analysis (QMEAN) server was used to determine the model quality (Benkert et al. 2008; Wiederstein and Sippl 2007). Moreover, the on the whole model quality was tested with Protein Structure Analysis (ProSA) software (<https://prosa.services.came.sbg.ac.at/prosa.php>). It offers a quality score, Z-score in comparison with all experimentally (X-ray crystallography as well as structural NMR) determined protein chains. These results signify the reliability of model that could be further used for the present docking studies.

Molecular docking and molecular dynamics studies

The molecular docking simulation was carried out to further investigate the binding mode of the top two most active compounds (**5r** and **5p**). The docking studies were performed using standard Glide (2019) molecular docking protocol implemented in Maestro molecular modelling suite of Schrödinger, LLC, New York, NY, 2019. The modelled structure of alpha glucosidase was prepared using default settings and the ligands were also prepared using the LigPrep module in a similar fashion as reported earlier (Malla et al. 2013). The docking was performed using Glide standard precision mode (Friesner et al. 2004, 2006; Halgren et al. 2004). The docked pose was further used for molecular dynamics simulation to assess the stability of the docked complex in time course using Desmond molecular dynamics program implemented in Schrödinger Maestro version 11.8.012. Molecular dynamics study was performed as reported earlier (Uniyal et al. 2020). Briefly, the docked complex of ligands **5r** and **5p** with the α -glucosidase was placed into orthorhombic box of optimal size according to the size of protein with an additional buffer length of 10 Å

in x, y and z direction and solvated explicitly with TIP3P waters. Further protocols for molecular dynamics were followed as per the procedure given in literature (Uniyal et al. 2020).

Results and discussion

Chemistry

The substituted benzimidazoles (**1a–f**), prepared from substituted *o*-phenylenediamines (Scheme 2), were converted to substituted benzimidazole alkynes (**2a–f**) by treating with propargyl bromide in presence of anhydrous K_2CO_3 in dimethylformamide (DMF) (Scheme 3). On the other hand, substituted indolines were reacted with bromoacetyl bromide in presence of anhydrous K_2CO_3 in dichloromethane (DCM) to give substituted 2-bromo-1-(indolin-1-yl) ethanones (**4a–c**). To produce the title compounds benzimidazole-1,2,3-triazole-indolines (**5a–r**), Cu(I)-catalysed Huisgen 1,3-dipolar cycloaddition reaction was performed between benzimidazole alkynes (**2a–f**) and 2-azido-1-(indolin-1-yl)ethanones formed in situ from 2-bromo-1-(indolin-1-yl)ethanone (**4a–c**) and sodium azide.

The structures of newly synthesized compounds were determined by analysing their spectroscopic data like Fourier

transform infrared (FTIR), one-dimensional (1D) and two-dimensional (2D) nuclear magnetic resonance (NMR) and high-resolution mass spectra (HRMS). The structure of compound **5d** (Fig. 1) was characterized through FTIR, 1H NMR, ^{13}C NMR, distortionless enhancement by polarization transfer (DEPT), correlation spectroscopy (COSY), heteronuclear single-quantum coherence (HSQC), heteronuclear multiple bond correlation (HMBC) and total correlation spectroscopy (TOCSY). FTIR data of **5d** display C=O absorption band at 1675 cm^{-1} . In 1H NMR, singlet at 8.06 ppm was assigned to triazole proton, i.e. C_{17} -H, two doublets at 7.94 and 7.14 ppm are due to C_{25} -H and C_{28} -H, respectively. Two doublets of doublets at 7.61 and 7.56 ppm were attributed to C_6 -H and C_3 -H. C_{26} -H and C_{27} -H gave triplets at 7.27 and 7.03 ppm, respectively. Singlets at 5.52 and 5.48 ppm arises due to C_{18} -H and C_{12} -H, respectively. The above peak assignment was supported by 2D NMR, i.e. COSY and TOCSY. COSY spectrum gave 1H - 1H correlation between C_1 -H/ C_2 -H and C_3 -H, C_1 -H/ C_2 -H and C_6 -H, C_{25} -H and C_{26} -H, C_{27} -H and C_{28} -H. The TOCSY spectrum suggested the correlation of C_3 -H with C_1 -H/ C_2 -H and C_6 -H and C_{27} -H with C_{25} -H, C_{26} -H and C_{28} -H.

In ^{13}C NMR spectrum of compound **5d**, twenty signals were found. DEPT indicated the presence of four secondary carbons, nine primary and tertiary carbons, thereby suggesting that the remaining seven carbons were quaternary

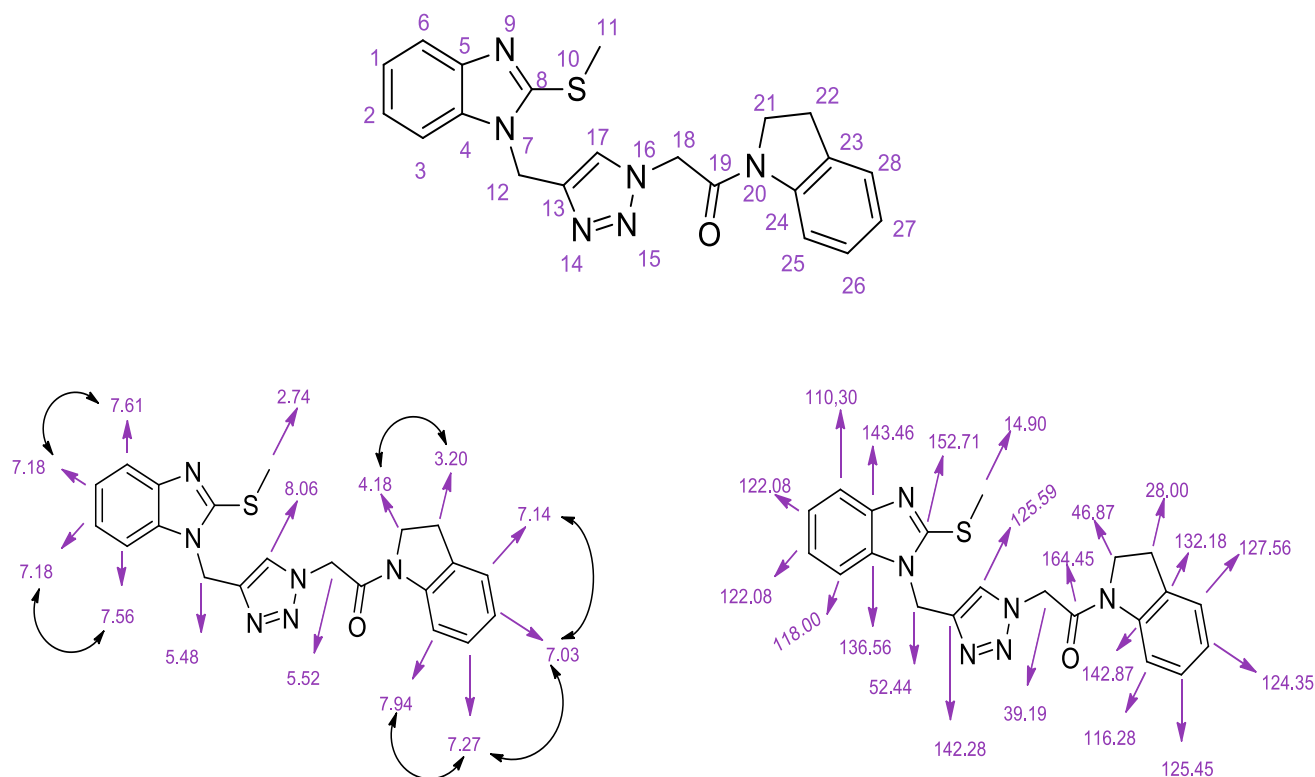


Fig. 1 1H and ^{13}C NMR chemical shift (δ , ppm) of compound **5d**

carbons. For the assignment of each carbon, HSQC and HMBC experiments were performed. HSQC gave the assignment of carbon signals at δ 127.56 (C_{28}), 125.59 (C_{17}), 125.45 (C_{26}), 122.08 (C_1/C_2), 118.00 (C_3), 116.28 (C_{25}), 110.12 (C_6), 52.44 (C_{12}), 46.87 (C_{21}), 39.19 (C_{18}), 28.00 (C_{22}), 14.90 (C_{11}), because the key correlation was as follows δ 8.06 $\rightarrow$ 125.59, 7.94 $\rightarrow$ 116.28, 7.61 $\rightarrow$ 110.30, 7.56 $\rightarrow$ 118.00, 7.27 $\rightarrow$ 125.45, 7.18 $\rightarrow$ 122.08, 7.14 $\rightarrow$ 127.56, 7.03 $\rightarrow$ 124.35, 5.52 $\rightarrow$ 39.19, 5.48 $\rightarrow$ 52.44, 4.18 $\rightarrow$ 46.87, 3.20 $\rightarrow$ 28.00 and 2.74 $\rightarrow$ 14.90.

The HMBC spectrum reflected the linkage involving hydrogen with other carbons through multiple bond correlation and help us to assign the quaternary carbon signals at 164.45 (C_{19}), 152.71 (C_8), 143.46 (C_5), 142.87 (C_{24}), 142.28 (C_{13}), 136.56 (C_4) and 132.18 (C_{23}). The above assignment was confirmed through HMBC correlation, i.e. δ 8.06 $\rightarrow$ 142.28; 7.94 $\rightarrow$ 132.18 and 125.45, 7.61 $\rightarrow$ 143.46 and 122.08; 7.56 $\rightarrow$ 143.46, 136.56 and 122.08; 7.27 $\rightarrow$ 142.87, 127.56 and 125.45; 7.18 $\rightarrow$ 143.46, 136.56, 118.00 and 110.30; 7.03 $\rightarrow$ 116.28 and 132.18; 5.52 $\rightarrow$ 164.45 and 125.59; 5.48 $\rightarrow$ 152.71, 142.28, 136.56 and 125.59.

In silico ADME properties

The drug-likeness properties of the newly synthesized compounds were calculated by online available Molinspiration property calculator (<https://www.molinspiration.com>) which gives information about the pharmacokinetic

parameters comprising of absorption, distribution, metabolism and excretion from the human body. The calculated parameters were reported in Table 2. The value of molecular aqua-phobicity is measured by the octanol–water partition coefficient (miLogP). All the newly synthesized compounds show miLogP value less than 5, these compounds may permeate lipid bilayers. The number of hydrogen bond acceptors (nON) was found less than or equal to 10, except in **5r** (nON = 11) and the number of hydrogen bond donors (nOHNH) in all the hybrid analogues was found to be less than 5. The synthesized compounds have molecular weight less than 500 Dalton which indicates their easy diffusion, absorption and transportation. The number of rotational bonds was found to be less than 10, thus demonstrating good flexibility. Topological polar surface area (TPSA) was observed in the range of 68.85 to 127.57 Å² which is less than 140 Å² thereby predicting good transport properties like intestinal absorption, blood barrier penetration, etc. All the newly synthesized compounds shown in Table 1 follow the ‘Lipinski rule of five’ except **5r** (one violation). There is one violation, which confirms the good drug-likeness of the rest of the synthesized compounds.

Pharmacology

Antimicrobial activity

The MIC values of the synthesized compounds ranges between 0.013 to 0.070 μ mol/mL for bacterial strains and

Table 2 In silico parameters of synthesized compounds

Comp. No	MiLogP	TPSA	Natoms	MW	nON	nOHNH	nviolation	Nrotb	Volume
5a	2.29	68.85	27	358.40	7	0	0	4	317.25
5b	3.09	68.85	29	386.46	7	0	0	4	350.37
5c	2.38	68.85	28	372.43	7	0	0	4	333.81
5d	3.21	68.85	29	404.50	7	0	0	5	351.94
5e	4.40	68.85	33	434.50	7	0	0	5	388.66
5f	3.26	81.75	33	435.49	8	0	0	5	384.50
5g	2.62	68.85	28	372.43	7	0	0	4	333.84
5h	3.42	68.85	30	400.49	7	0	0	4	366.96
5i	2.71	68.85	29	386.46	7	0	0	4	350.40
5j	3.54	68.85	30	418.53	7	0	0	5	368.53
5k	4.73	68.85	34	448.53	7	0	0	5	405.25
5l	3.59	81.75	34	449.52	8	0	0	5	401.09
5m	2.22	114.68	30	403.40	10	0	0	5	340.58
5n	3.02	114.68	32	431.46	10	0	0	5	373.71
5o	2.31	11.68	31	417.43	10	0	0	5	357.14
5p	3.14	114.68	32	449.50	10	0	0	6	375.27
5q	4.34	114.68	36	479.50	10	0	0	6	411.99
5r	3.19	127.57	36	480.49	11	0	1	6	407.84

MW molecular weight, miLogP calculated partition coefficient, nRotB number of rotatable bonds, TPSA topological polar surface area

0.026 to 0.070 $\mu\text{mol/mL}$ for fungal strains, respectively (Table 2). Analysis of the antimicrobial activity result from Table 3 showed that compound **5r** showed excellent activity against bacteria, i.e. *S. aureus* ($\text{MIC}=0.013 \mu\text{mol/mL}$), *E. coli* and *S. epidermidis* ($\text{MIC}=0.026 \mu\text{mol/mL}$) and compound **5o** showed good activity against bacteria, i.e. *E. coli* and *S. epidermidis* ($\text{MIC}=0.030 \mu\text{mol/mL}$) as compared to reference drug Norfloxacin. Compound **5r** was found to possess promising activity against *A. niger* and *C. albicans* with MIC value of 0.0260 $\mu\text{mol/mL}$ as compared to reference drug Fluconazole.

- In benzimidazole-1,2,3-triazole-indoline derivatives **5f**, **5l** and **5r** having pyridinyl moiety on benzimidazole, the order of activity is **5r** (NO_2) > **5f** $\approx$ **5l** (H) against *S. aureus*, *E. coli*, *S. epidermidis*, *A. niger* and *C. albicans*. Interestingly, **5r** displayed highest activity against all microbes except *B. subtilis*, even better than standard drugs—Norfloxacin and Fluconazole. Further, in thio-methyl derivatives (**5d**, **5j** and **5p**) substitution of nitro group on indoline again performed better activity against all bacterial and fungal strains and in methyl derivatives (**5c**, **5i** and **5o**), **5o** with nitro substitution was more active against *E. coli* and *S. epidermidis* as compared to reference drug—Norfloxacin.
- In benzimidazole-1,2,3-triazole-indoline derivatives **5e**, **5k** and **5q** having phenyl moiety on benzimidazole, there is comparable activity against all strains except for *S. aureus* where the order is **5e** (H) $\approx$ **5q** (NO_2) > **5k** (H).

- Overall, nitro and pyridinyl containing derivatives exhibited better antimicrobial potential among the synthesized compounds.

α -Glucosidase inhibition

The newly synthesized compounds were evaluated for in vitro inhibition of α -glucosidase using Acarbose as a standard drug. The IC_{50} values of compounds are presented in Table 4. The predominance of tested compounds showed significant α -glucosidase inhibitory activity with IC_{50} values range of 0.015 ± 0.0003 – $0.081 \pm 0.002 \mu\text{mol/mL}$ as compared to standard drug Acarbose ($\text{IC}_{50}=0.019 \pm 0.0007 \mu\text{mol/mL}$). Compounds **5p** ($\text{IC}_{50}=0.015 \pm 0.0003 \mu\text{mol/mL}$) and **5r** ($\text{IC}_{50}=0.018 \pm 0.0008 \mu\text{mol/mL}$) showed excellent potency against α -glucosidase as compared to standard drug.

Structure–activity relationship

The results of the α -glucosidase inhibition demonstrated some interesting facts about the SAR of the synthesized benzimidazole-1,2,3-triazoles-indolines (Fig. 2). The pharmacological profile of molecules is strongly affected by the substituents present on benzimidazole and indoline ring. Further, while comparing the α -glucosidase inhibition data of our previously reported work with the current manuscript

Table 3 Antimicrobial activity of synthesized compounds in terms of MIC ($\mu\text{mol/mL}$)

Compound	<i>S. aureus</i>	<i>E. coli</i>	<i>B. subtilis</i>	<i>S. epidermidis</i>	<i>A. niger</i>	<i>C. albicans</i>
5a	0.035	0.070	0.070	0.070	0.070	0.070
5b	0.065	0.065	0.065	0.065	0.065	0.065
5c	0.067	0.067	0.067	0.067	0.067	0.067
5d	0.062	0.062	0.062	0.062	0.062	0.062
5e	0.029	0.058	0.058	0.058	0.058	0.058
5f	0.029	0.057	0.057	0.057	0.057	0.057
5g	0.034	0.067	0.067	0.067	0.067	0.067
5h	0.031	0.063	0.063	0.063	0.063	0.063
5i	0.065	0.065	0.065	0.065	0.065	0.065
5j	0.060	0.060	0.060	0.060	0.060	0.060
5k	0.056	0.056	0.056	0.056	0.056	0.056
5l	0.028	0.056	0.056	0.056	0.056	0.056
5m	0.031	0.062	0.062	0.062	0.062	0.062
5n	0.029	0.058	0.058	0.058	0.058	0.058
5o	0.060	0.030	0.060	0.030	0.060	0.060
5p	0.029	0.056	0.056	0.056	0.056	0.056
5q	0.026	0.052	0.052	0.052	0.052	0.052
5r	0.013	0.026	0.052	0.026	0.026	0.026
Norfloxacin	0.020	0.039	0.039			
Fluconazole					0.041	0.020

(as shown in Scheme 1, Deswal et al. 2020), it observed that on replacement of phenyl ring with the indoline ring led to significant enhancement of activity. The notable highlights are summarized below:

- In indoline derivatives (**5a–f**), benzimidazoles having pyridinyl group at R² position were more active followed by thiomethyl, phenyl, methyl and hydrogen against α -glucosidase inhibition.
- In methyl indoline derivatives (**5 g–l**), compound **5l** active with pyridinyl group at R² position and same trend was followed as in indoline derivatives.
- In nitro indoline derivatives (**5 m–r**), compound **5p** was most active in the entire series with thiomethyl group and compound **5r** was also more active against α -glucosidase as compared to standard drug Acarbose.

Table 4 α -Glucosidase inhibition of synthesized compounds (**5a–r**) in terms of IC₅₀ in $\mu\text{mol/mL}$ with standard deviation

Compound	α -glucosidase inhibition (IC ₅₀ in $\mu\text{mol/mL}$)	Compound	α -glucosidase inhibition (IC ₅₀ in $\mu\text{mol/c}$)
5a	0.081 $\pm$ 0.002	5j	0.032 $\pm$ 0.0007
5b	0.048 $\pm$ 0.0009	5l	0.035 $\pm$ 0.0009
5c	0.056 $\pm$ 0.0015	5l	0.023 $\pm$ 0.0008
5d	0.037 $\pm$ 0.0011	5m	0.036 $\pm$ 0.001
5e	0.046 $\pm$ 0.0014	5n	0.029 $\pm$ 0.0016
5f	0.030 $\pm$ 0.0008	5o	0.034 $\pm$ 0.0014
5g	0.059 $\pm$ 0.0011	5p	0.015 $\pm$ 0.0003
5h	0.049 $\pm$ 0.0018	5q	0.027 $\pm$ 0.0006
5i	0.043 $\pm$ 0.0014	5r	0.018 $\pm$ 0.0008
Acarbose	0.019 $\pm$ 0.0007		

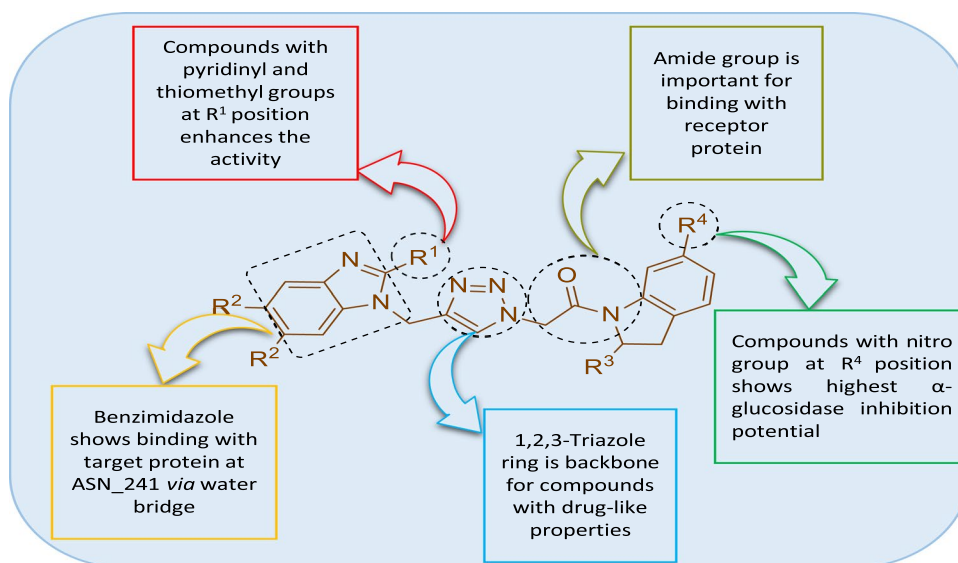
- Overall, compounds with nitro group at R⁴ position and pyridinyl or thiomethyl group at R² position exhibited better α -glucosidase inhibition potential among the synthesized compounds.

Molecular docking and molecular dynamics studies

Molecular docking studies of the two most active compounds (**5p** and **5r**) were performed to investigate the binding interactions of these compounds in the active site of yeast α -glucosidase. Both the ligands docked well into the active site of the enzyme and the docking pose obtained were further subjected to molecular dynamics simulation for a duration of 25 ns as described in the methodology section.

Figure 3 shows the molecular dynamics analysis data for compound **5p**. The value of RMSD was between 0.9 to 2.5 Å (Fig. 3a, RMSD plot) and it confirms stability of complex of protein with compound **5p** during the whole simulation period. RMSF value (0.5 to 2.7 Å) of each amino acid residues gave information about the protein flexibility during the simulation (Fig. 3b). The α -glucosidase active site contacts of compound **5p** were quantified and is given in Fig. 3c which indicates that the ligand **5p** interacts well with the binding site of α -glucosidase mainly via hydrogen bonding with ASN_241 and GLN_350, by hydrophobic interaction PHE_157 and PHE_158, PHE_300 and HIS_239. Water bridges were also observed with residues ASN_241, GLU_276 and ARG_312. RMSF sketch of compound **5p** (Fig. 3d) shows that the ligand mainly remained anchored to the binding site residues and showed a RMSF value of 1–2.6 Å. Figure 3e gives the quantitative assessment of ligand protein interaction during the time course of MD simulation. It is observed that the ligand **5p** interacted with the PHE_157, ASN_241 and GLU_276 throughout

Fig. 2 SAR of benzimidazole-1,2,3-triazole-indoline derivatives



the simulated time of 25 ns. The plot in Fig. 3f displays hydrogen bonds, ionic as well as hydrophobic interaction and water bridges as the timeline of contacts between ligand and protein all the way through the 25 ns of simulation time. At the top section, the total number of specific interactions between protein and ligand is shown while at the bottom

slice residue level contacts of the ligand are pictured. On the whole, the compound **5p** affixed well in the binding site of the yeast α -glucosidase.

Similarly, Fig. 4 shows the molecular dynamics analysis data for compound **5r**. The root mean square deviation (RMSD) plot (Fig. 4a) exhibits that the ligand and protein

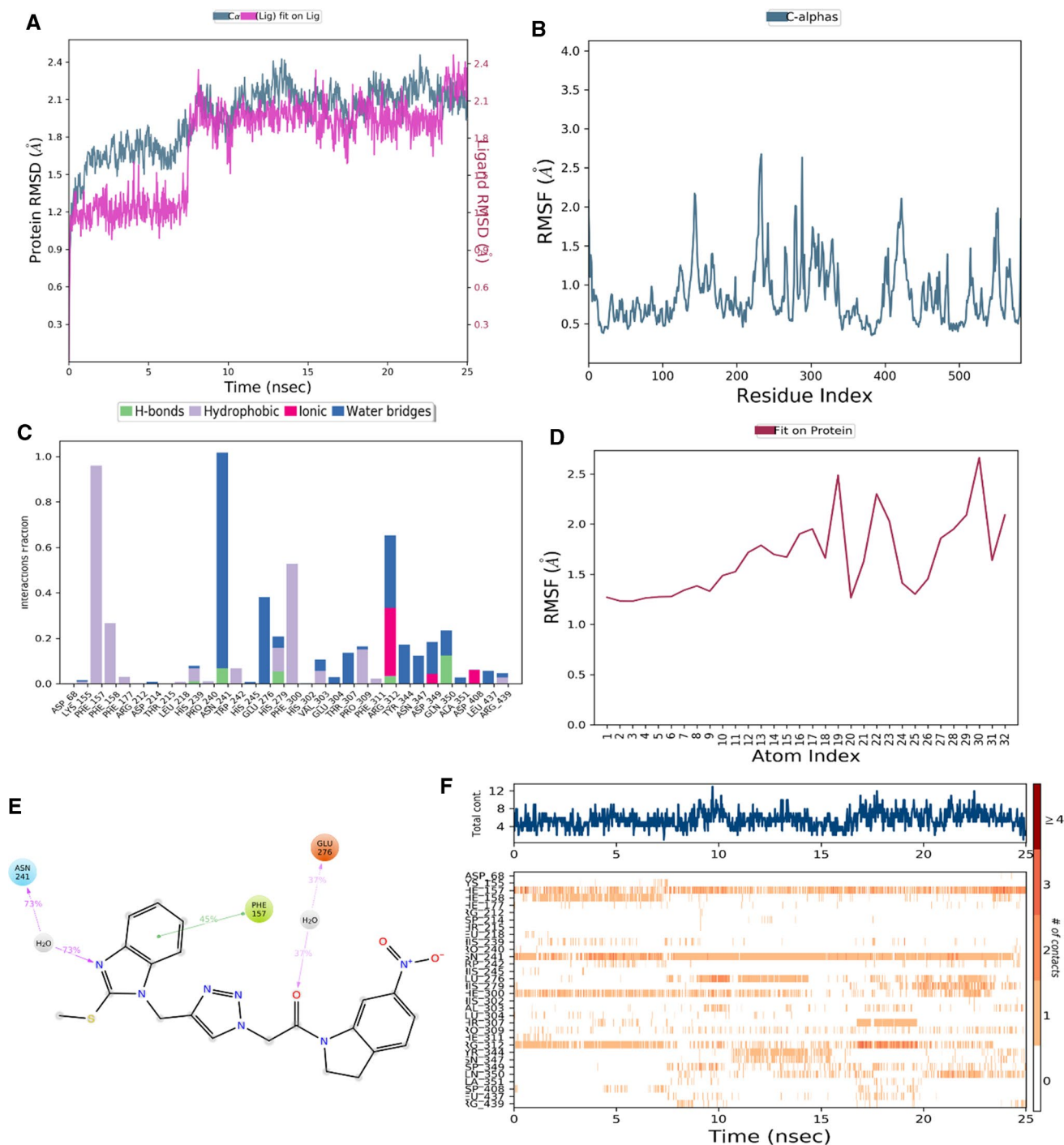


Fig. 3 Molecular dynamics trajectory analysis of ligand **5p** docked to α -glucosidase. **a** RMSD of the protein and ligand with respect to the first frame **b** Protein RMSF **c** Protein–ligand contacts histogram **d** Ligand RMSF **e** Ligand–protein contacts **f** Protein–ligand contacts

remain stable throughout the simulation period and the RMSD value was between 1.2 to 2.7 Å for the protein and 0.3 to 2.9 Å for the ligand, signifying a steady ligand–protein complex. The root mean square fluctuation (RMSF) calculations of amino acids checked the protein flexibility during simulations (Fig. 4b). The RMSF value for the entire protein was between 0.4 to 3.9 Å. Different interactions between the ligand **5r** and active site amino acids residues were quantified and are given in Fig. 4c, which indicates that the ligand

5r interacts profoundly in the binding site of α -glucosidase mainly via hydrogen bonding with LYS_155, by hydrophobic interaction PHE_157 and PHE_158, PHE_177 and HIS_239. Results of RMSF plot for compound **5r** (Fig. 4d, RMSF value of 2–4 Å) show steady binding with the amino acids largely. Figure 4e gives the quantitative assessment of ligand protein interaction during the time course of MD simulation. It is observed that the ligand **5r** interacted with the PHE_177, PHE_157 and LYS_155 throughout the simulated

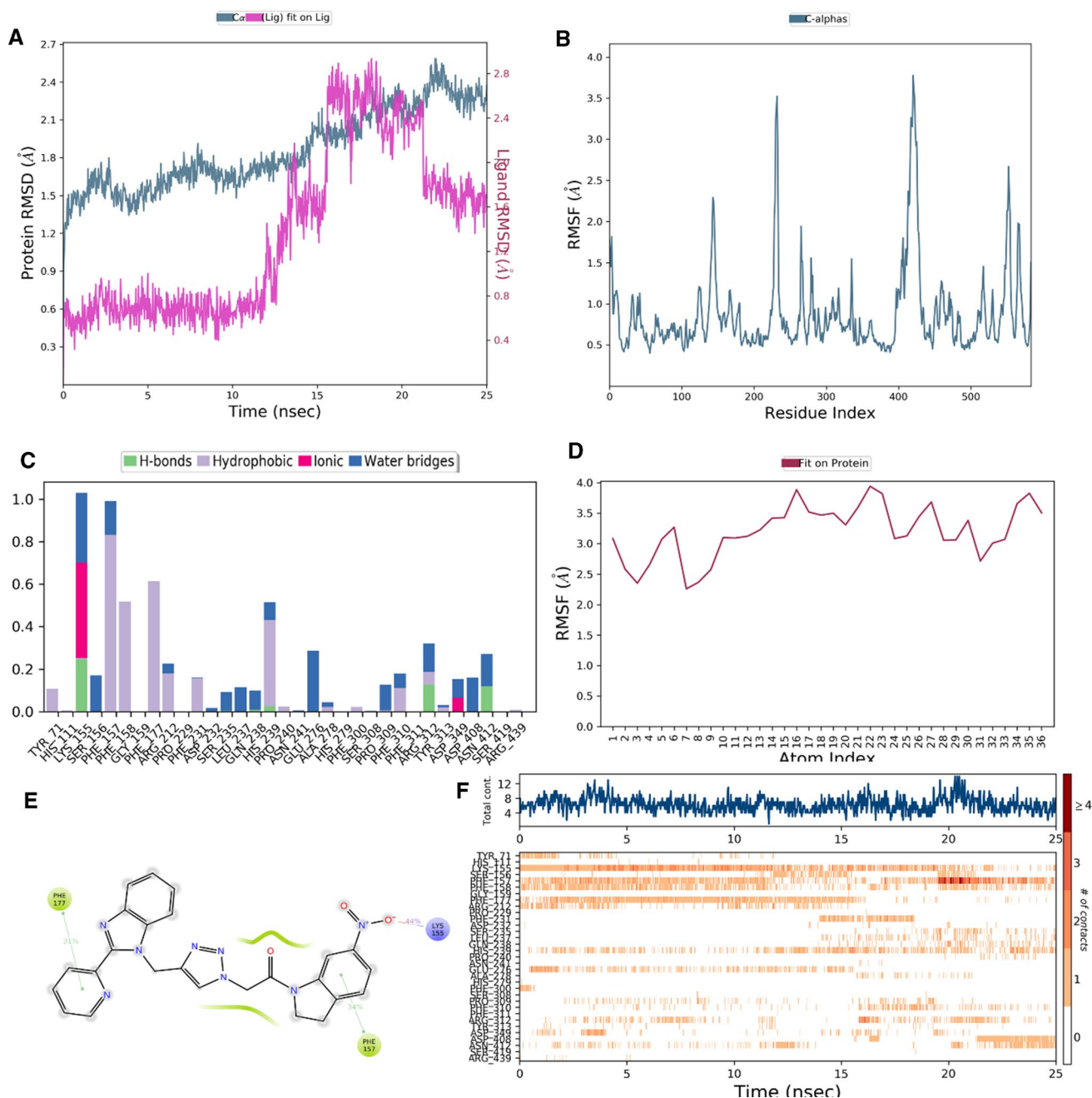


Fig. 4 Molecular dynamics trajectory analysis of ligand **5r** docked to α -glucosidase. **a** RMSD of the protein and ligand with respect to the first frame **b** Protein RMSF **c** Protein–ligand contacts histogram **d** Ligand RMSF **e** Ligand–protein contacts **f** Protein–ligand contacts

time of 25 ns. The plot in Fig. 4f displays hydrogen bonds, ionic interaction, hydrophobic interaction and water bridges as the timeline of interactions between ligand and protein all the way through the 25 ns of simulation time. At the top section, the total number of specific interactions between protein and ligand is shown while at the bottom slice residue level contacts of the ligand are pictured. On the whole, the compound **5r** anchored well in the active site of the yeast α -glucosidase.

Conclusions

In conclusion, click chemistry-inspired approach led to synthesis of eighteen benzimidazole-1,2,3-triazole-indolines showing good ADME properties, which were evaluated for their in vitro antimicrobial and α -glucosidase inhibition. Compound **5o** displayed promising activity against *E. coli*, while compound **5r** showed good inhibition against all the tested strains except *B. subtilis*. Further, compounds **5p** and **5r** exhibit remarkable inhibitory activity against the α -glucosidase enzyme. Molecular docking and molecular dynamics studies of **5p** and **5r** revealed binding conformation responsible for α -glucosidase inhibition potential. It also unveils that **5p** interacted well with the PHE_157, ASN_241 and GLU_276 and **5r** interacted with the PHE_177, PHE_157 and LYS_155 residues.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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